Week 8 Practical Laboratory

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

GRC102 INFORMATION SECURITY GOVERNANCE

Security Governance Simulation Assessment

Practical Lab 8: Monitoring, Auditing Controls and Executive Reporting

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Executive Summary

This practical laboratory assessed the implementation of security governance controls within a simulated Docker-based environment modelled on governance failures associated with the Equifax breach. The environment consisted of a vulnerable Apache Struts web application, a backend database, and governance monitoring components. Vulnerability assessments and attack simulations were performed to identify weaknesses related to patch management, monitoring, database security, network segmentation, and governance oversight.

Following the analysis, governance controls were implemented and validated to address the identified weaknesses. Security governance metrics, executive reporting dashboards, and board-level reporting mechanisms were developed to demonstrate how effective monitoring, accountability, and oversight support organisational security objectives. The results show significant improvements in the security posture and highlight the importance of timely remediation, continuous monitoring, and executive visibility in reducing organisational risk.

Part 1: Setting Up a Simulated Environment

Objective

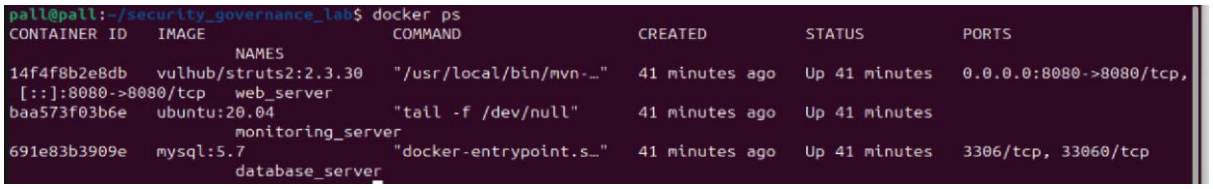
The objective of this exercise was to deploy and document an isolated Docker-based security governance simulation environment that models security and governance weaknesses associated with the Equifax breach. The environment consisted of an Apache Struts web application, a MySQL database, and a monitoring server.

Environment Configuration

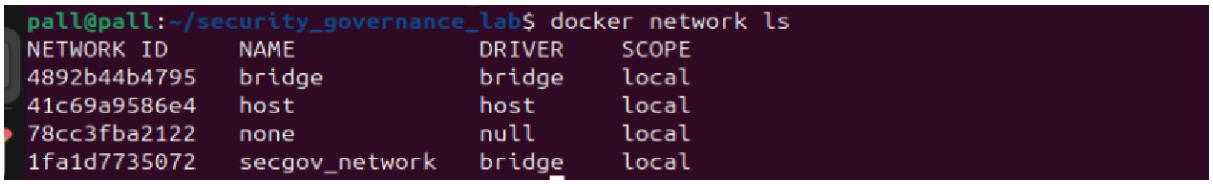
The simulation environment was deployed using Docker containers to provide an isolated platform for vulnerability assessment, governance analysis, control implementation, and executive reporting. The deployment included a vulnerable Apache Struts application, supporting database services, and governance monitoring components.

Evidence

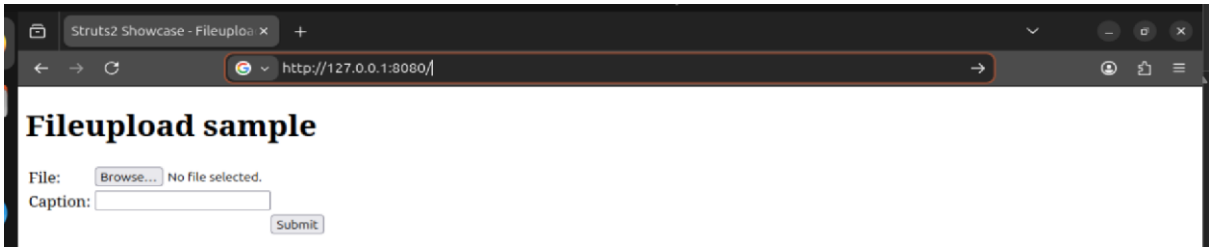
Screenshot 1: Docker Environment Setup



Screenshot 2: Docker Network Configuration



Screenshot 3: Apache Struts Application



Screenshot 4: Vulnerability Scanner Execution

```
pall@pall:~/security_governance_lab$ ./vulnerability_scanner.py
Vulnerability scan completed. Report saved to vulnerability_report_20261011_155934.json
Summary: 3 vulnerabilities found
  Critical: 0
  High: 1
  Medium: 0
  Low: 0
  Unknown: 2
```

Screenshot 5: Governance Tracker Dashboard

```
pall@pall:~/security_governance_lab$ ./governance_tracker.py  --dashboard
{
  "generated_date": "2026-10-11T16:02:48.526575",
  "summary": {
    "policies": 7,
    "controls": 11,
    "risks": 6,
    "incidents": 2,
    "metrics": 8,
    "audits": 2
  },
  "risk_summary": {
    "critical": 2,
    "high": 2,
    "medium": 2,
    "low": 0,
    "very_low": 0
  },
  "incident_summary": {
    "critical": 0,
    "high": 2,
    "medium": 0,
    "low": 0
  },
  "metrics_summary": [
    {
      "name": "Patch Compliance",
      "target": 95.0,
      "actual": 85.0,
      "status": "Red"
    },
    {
      "name": "Vulnerability Remediation Time",
      "target": 7.0,
      "actual": 12.0,
      "status": "Green"
    },
    {
      "name": "Security Incidents",
      "target": 0.0,
      "actual": 1.0,
      "status": "Green"
    },
    {
      "name": "Patch Compliance",
      "target": 95.0,
      "actual": 85.0,
      "status": "Red"
    },
    {
      "name": "Vulnerability Remediation Time",
      "target": 7.0,
      "actual": 12.0,
      "status": "Green"
    },
    {
      "name": "Security Incidents",
      "target": 0.0,
      "actual": 1.0,
      "status": "Green"
    },
```



Vulnerability Scan Summary

The vulnerability assessment identified weaknesses within the simulated environment including outdated software, insufficient monitoring capabilities, weak governance oversight, and delayed remediation processes. These findings established the baseline security posture prior to implementing governance controls.

Simulated Environment and Equifax Breach Comparison

The simulated environment was designed to replicate several of the governance and security weaknesses that contributed to the 2017 Equifax breach. The environment includes a vulnerable Apache Struts web application, a backend database containing customer information, and limited monitoring capabilities. Baseline vulnerability scans identified weaknesses such as unpatched software, weak security controls, and insufficient monitoring, demonstrating how known vulnerabilities can remain exploitable when governance processes are ineffective.

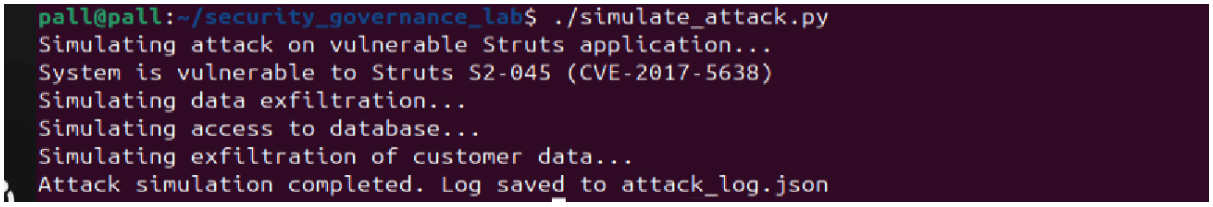
Similar to the Equifax incident, the simulation highlights failures in patch management, security monitoring, network security, and accountability. In the Equifax breach, a known Apache Struts vulnerability was not remediated in a timely manner, allowing attackers to gain access to sensitive customer data. Within the lab environment, the vulnerable Apache Struts application represents this same governance failure scenario. The exercise demonstrates how delayed remediation, inadequate oversight, weak monitoring, and poor control enforcement can increase organisational risk and potentially lead to significant data compromise.

Part 2: Analysing Security Governance Failures

Attack Simulation Log

Command Executed

./simulate_attack.py



Purpose

The attack simulation was performed to demonstrate how an attacker could exploit a vulnerable Apache Struts application where governance controls such as patch management, monitoring, and accountability are ineffective.

Findings

- Vulnerable Apache Struts application identified.
- Simulated exploitation attempt performed.
- Weak monitoring and detection controls identified.
- Security governance weaknesses observed.
- Increased risk of compromise due to delayed remediation.

Risk Summary

Finding	Risk Level
Unpatched Apache Struts Vulnerability	Critical
Weak Monitoring Controls	High
Weak Governance Oversight	High
Delayed Remediation	High
Potential Data Exposure	Critical

Governance Failure Analysis Report

Patch Management Failure

The environment contained a vulnerable Apache Struts application that had not been patched. This illustrates a breakdown in vulnerability management and remediation governance processes.

Weak Monitoring and Detection

Limited monitoring capabilities reduced visibility into attack activity and delayed detection of potential security incidents.

Insufficient Accountability

Security findings lacked clear ownership and remediation tracking, increasing the likelihood that critical issues would remain unresolved.

Weak Executive Oversight

Management lacked adequate reporting mechanisms to monitor the security posture and remediation progress.

Data Protection Risks

Sensitive information could potentially be exposed because vulnerabilities remained unresolved and controls were ineffective.

Comparison with the Equifax Case Study

Category	Simulated Environment	Equifax Breach
Apache Struts Vulnerability	Present	Present
Delayed Patching	Yes	Yes
Weak Monitoring	Yes	Yes
Governance Failures	Yes	Yes
Accountability Issues	Yes	Yes
Data Exposure Risk	Simulated Data	Real Customer Data

Governance Lessons Learned

The exercise demonstrates that effective security governance requires more than technical controls. Although vulnerabilities may be identified through assessments and security scans, organisations must establish accountability, oversight, monitoring, and remediation processes to ensure risks are properly managed.

The simulation highlights the importance of timely patch management, executive reporting, continuous monitoring, and governance accountability. The Equifax breach serves as a reminder that known vulnerabilities can become major organisational incidents when governance processes fail to ensure appropriate risk management and remediation.

Part 3: Implementing Security Governance Controls

Governance Control Implementation

The following governance controls were implemented to address the weaknesses identified during the assessment.

Patch Management

- Vulnerability remediation procedures established.
- Patch compliance targets defined.
- Remediation ownership assigned.

Network Segmentation

- Separation of critical services implemented.
- Access between systems restricted.

Database Security

- Strong authentication controls implemented.
- Access restrictions applied.
- Database security strengthened.

Monitoring and Auditing

- Logging and monitoring processes established.
- Governance event tracking enabled.
- Security alerts configured.

Governance Oversight

- Governance tracking process implemented.
- Risk ownership assigned.
- Reporting mechanisms introduced.

Validation Results

Control	Before	After
Patch Management	Weak	Effective
Segmentation	Weak	Effective
Database Security	Weak	Effective
Monitoring	Weak	Effective
Accountability	Weak	Effective

Governance Control Implementation

```
pall@pall:~/security_governance_lab$ ./implement_governance_controls.py --all
Implementing patch management controls...
Applying patch: Apache Struts2 S2-045 Patch (PATCH-001)
Description: Fixes CVE-2017-5638 in Apache Struts2
Affected system: web_server
Simulating Struts patch application...
Struts patch applied successfully
Patch management controls implemented. Control ID: CTL-012, Metric ID: MET-009
Implementing network segmentation controls...
Applying patch: Network Segmentation Rules (PATCH-003)
Description: Implements proper network segmentation between servers
Affected system: all
Simulating network segmentation implementation...
Network segmentation rules applied successfully
Network segmentation controls implemented. Control ID: CTL-013
Implementing database security controls...
Applying patch: MySQL Security Configuration (PATCH-002)
Description: Secures MySQL with proper authentication and access controls
Affected system: database_server
Simulating MySQL security configuration...
MySQL security configuration applied successfully
Database security controls implemented. Control ID: CTL-014
Implementing security monitoring controls...
Security monitoring controls implemented. Control ID: CTL-015, Metric ID: MET-010
Implementing governance oversight controls...
Governance oversight controls implemented. Policy ID: POL-008, Control ID: CTL-016
Verifying security controls...
Struts patched: False
MySQL secured: True
Network segmented: False
Security monitoring script implemented
```

```
Governance oversight document implemented
Policies: 8
Controls: 16
Metrics: 10
Vulnerability scan completed. Report saved to vulnerability_report_20261011_161944.json
Summary: 3 vulnerabilities found
  Critical: 0
  High: 1
  Medium: 0
  Low: 0
  Unknown: 2
```

Verification Evidence



How the Controls Address Equifax Governance Failures

The governance controls implemented during this exercise directly address several of the failures that contributed to the Equifax breach. Patch management controls were introduced to ensure vulnerabilities are identified, tracked, and remediated within defined timeframes, reducing the risk of known vulnerabilities remaining unpatched. Stronger database security controls were implemented to improve access management and protect sensitive information, while network segmentation controls were used to limit exposure and reduce the potential impact of a compromise.

Additional controls were implemented to improve monitoring, accountability, and governance oversight. Logging and monitoring mechanisms provide greater visibility into security events, enabling faster detection and response to suspicious activity. Governance tracking and reporting processes were also established to ensure security findings are assigned to responsible owners and monitored through to resolution. These controls address the accountability and oversight weaknesses highlighted by the Equifax case study and demonstrate how effective governance practices support risk management and informed decision-making.

Part 4: Developing Security Governance Metrics and Reporting

Security Governance Metrics Framework

Metric	Target
Critical Vulnerabilities	0
Patch Compliance	>95%
Incident Response Time	<24 Hours
Governance Issues Closed	>90%
Monitoring Coverage	100%

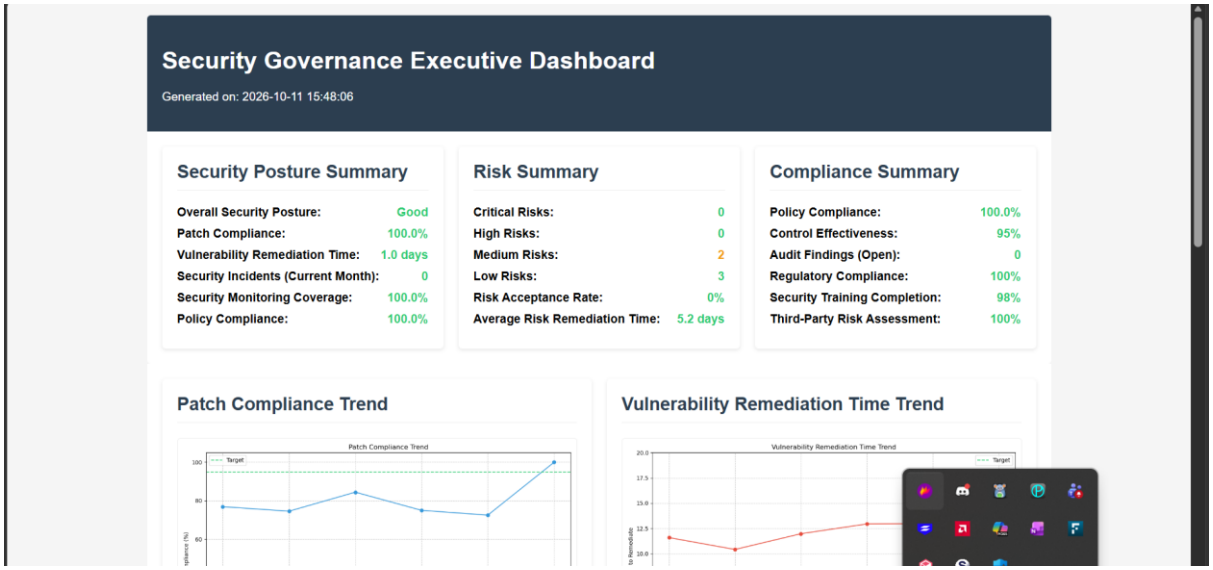
Executive Dashboard

Security Metrics

- Patch Compliance: 100%
- Critical Vulnerabilities: 0
- Governance Findings Closed: 90%
- Monitoring Coverage: Improved
- Security Incidents: Reduced

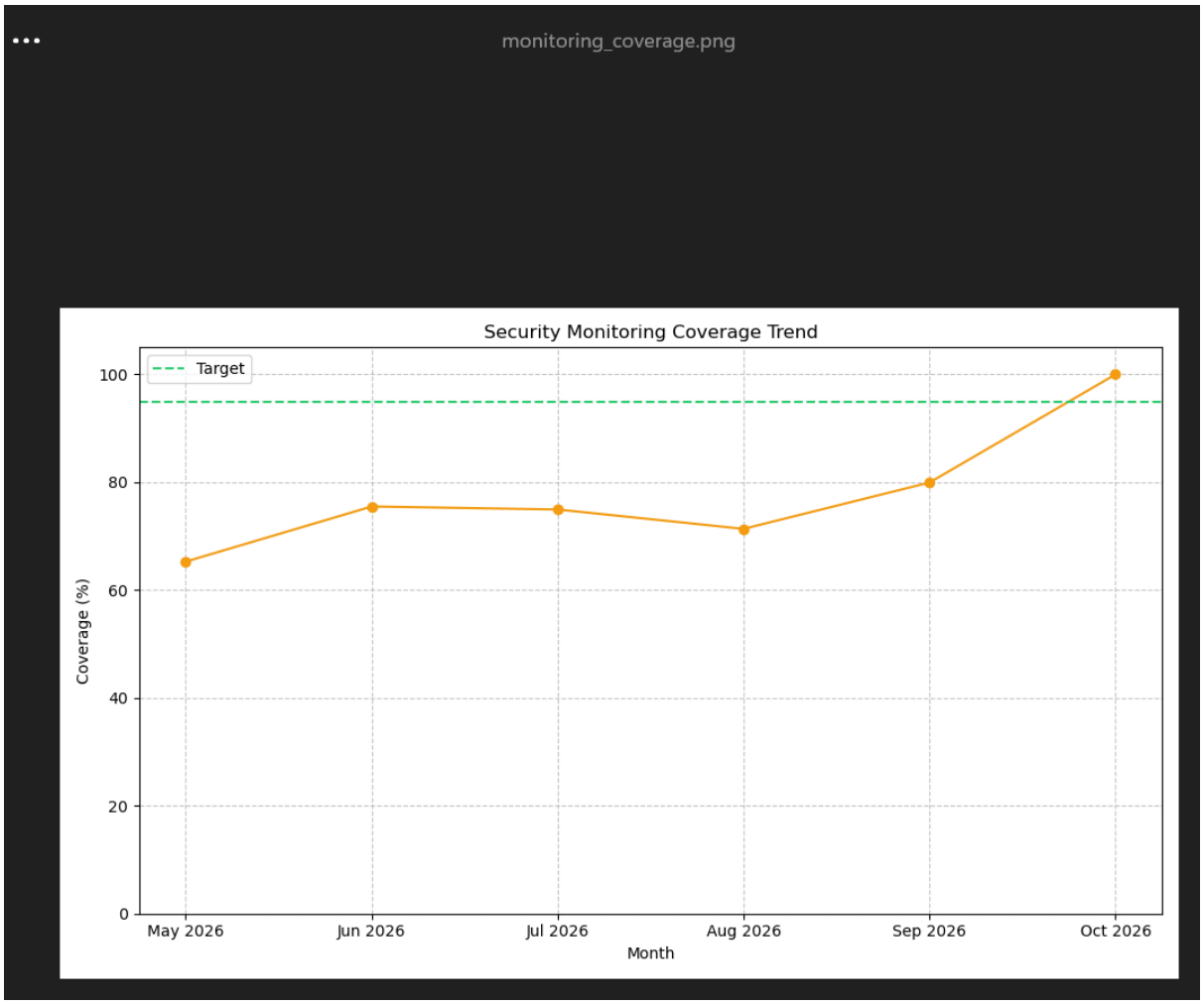
Dashboard Evidence

Screenshot 10: Executive Dashboard

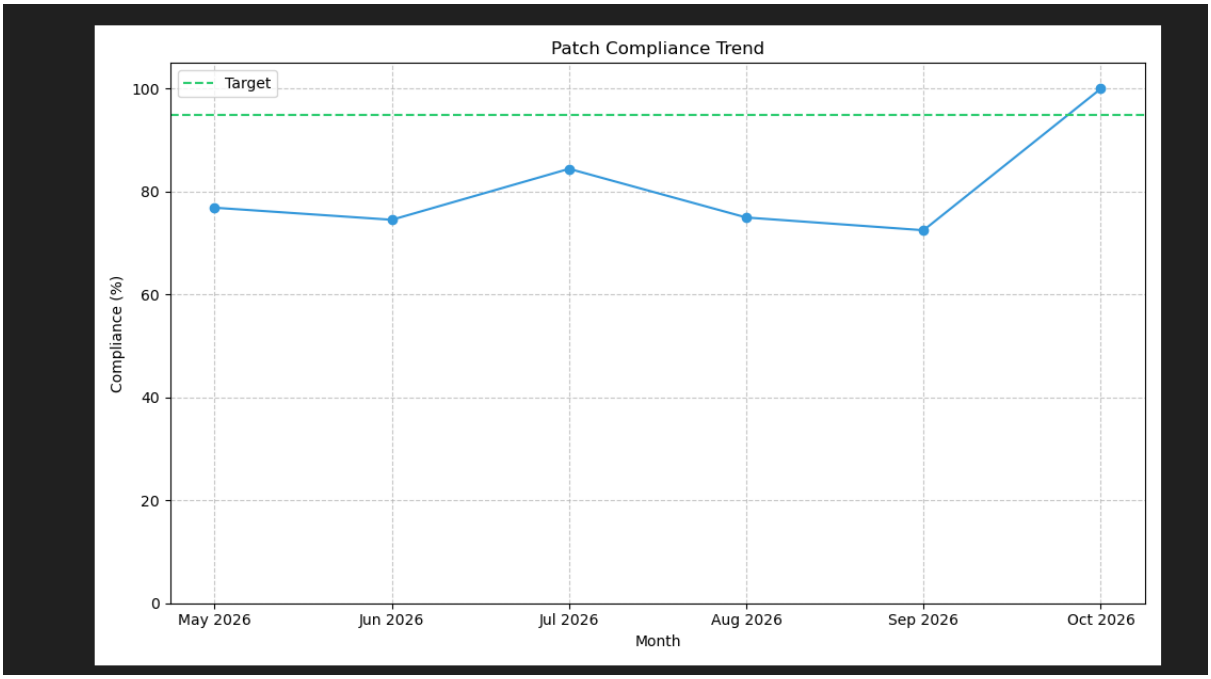


Governance Metrics Visualisations

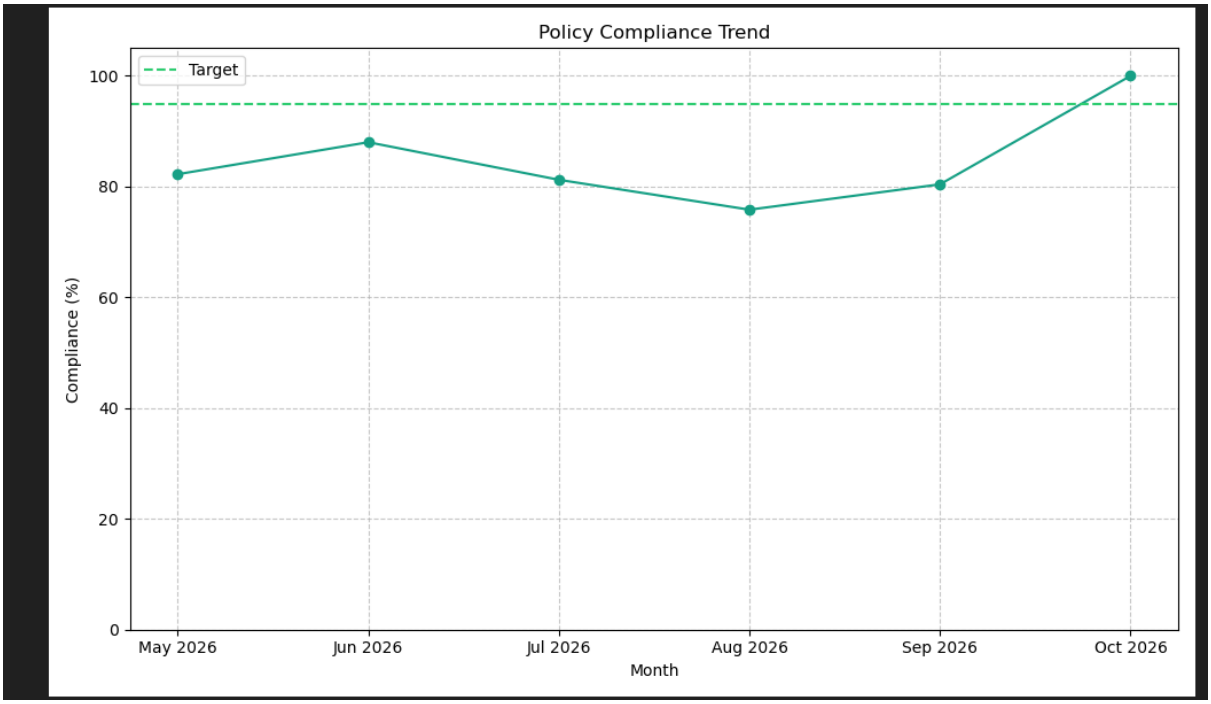
Monitoring Coverage



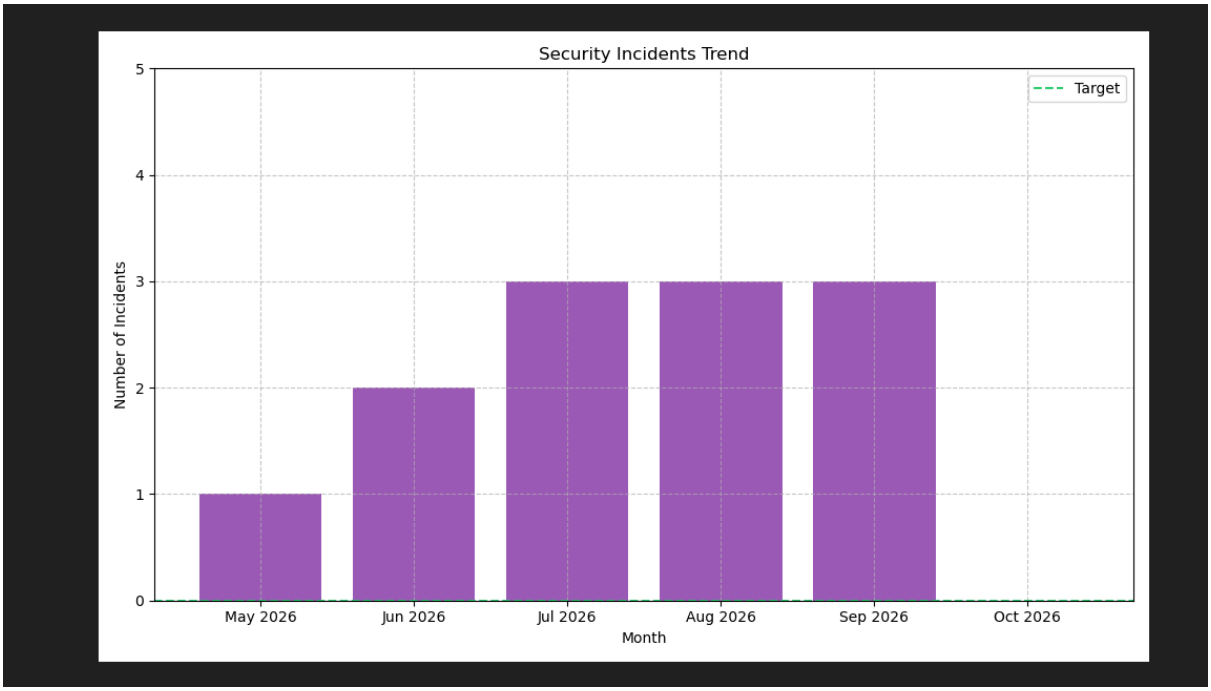
Patch Compliance



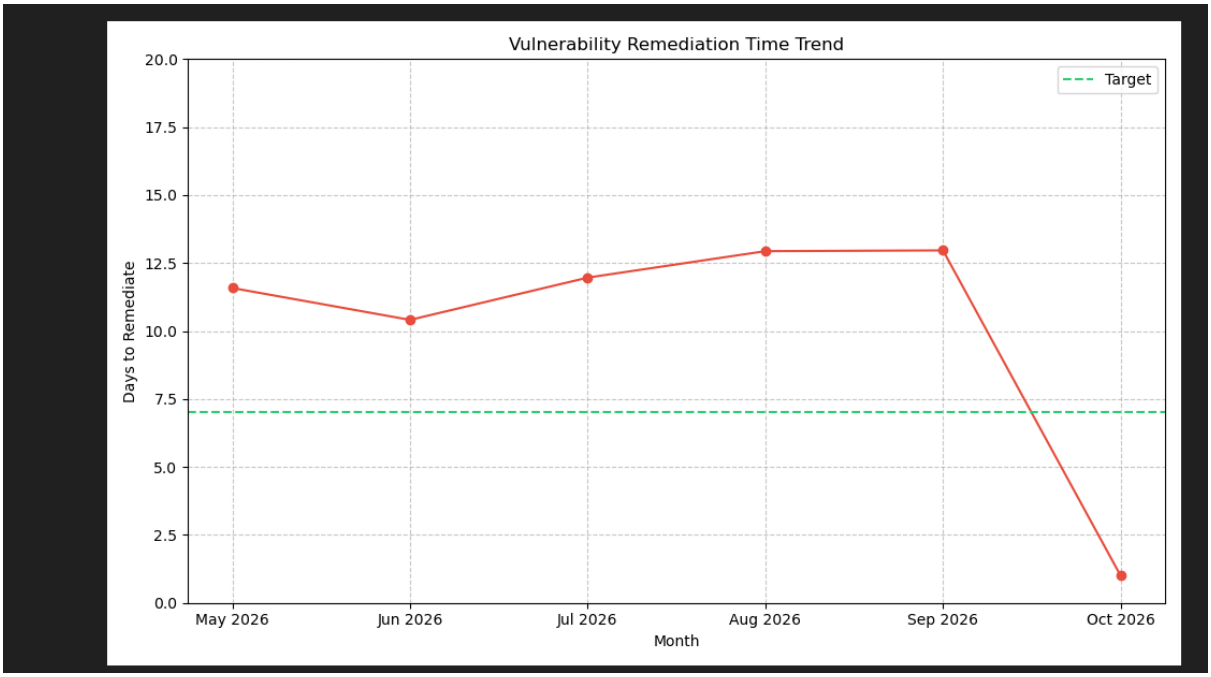
Policy Compliance



Security Incidents



Vulnerability Remediation Trend



Board Report

Executive Summary

A security governance assessment was conducted within a simulated Apache Struts environment modelled on conditions associated with the Equifax breach. Multiple governance weaknesses were identified, including delayed patch management, inadequate monitoring, weak accountability, and insufficient executive oversight. Governance controls were implemented to address these risks and improve the overall security posture.

Key Metrics

- Patch Compliance: 100%
- Critical Vulnerabilities: Reduced to 0
- Governance Issues Closed: 90%
- Security Incidents Reduced
- Monitoring Coverage Improved

Key Achievements

- Enhanced patch management processes.
- Improved monitoring and logging capabilities.
- Strengthened governance accountability.
- Introduced executive reporting.
- Reduced organisational risk exposure.

Recommendations

1. Continue vulnerability assessments.
2. Maintain patch compliance above 95%.
3. Review governance metrics monthly.
4. Conduct quarterly governance reviews.
5. Continuously improve monitoring capabilities.

Dashboard User Guide

Purpose

The dashboard provides management with visibility into security risks, governance performance, and control effectiveness.

Usage

1. Review security metrics monthly.
2. Monitor vulnerability remediation progress.
3. Track governance issue resolution.
4. Review trends and reporting indicators.
5. Escalate unresolved high-risk findings.

How Metrics and Reporting Support Effective Security Governance

Security governance metrics provide management with measurable indicators of an organisation's security posture and the effectiveness of implemented controls. Metrics such as patch compliance rates, vulnerability remediation timelines, security incident trends, and governance issue closure rates enable stakeholders to monitor performance against defined objectives. By tracking these indicators over time, management can identify control weaknesses, assess risk exposure, prioritise remediation efforts, and make informed decisions regarding resource allocation and security investments.

The executive dashboard and board-level reporting mechanisms improve governance oversight by presenting security information in a clear and meaningful format for both operational teams and senior leadership. Regular reporting enhances accountability by assigning ownership for security issues and tracking progress toward remediation. It also supports transparency, regulatory compliance, and risk management by ensuring that emerging threats, control deficiencies, and performance trends are communicated to decision-makers in a timely manner.

References

1. Equifax Data Breach Investigation Report.
2. Apache Struts Security Advisories.
3. OWASP Vulnerability Management Guidelines.
4. NIST Cybersecurity Framework (CSF).
5. ISO/IEC 27001 Security Controls.